

Aayush

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Profile Summary

Aspiring AI/ML Research Engineer (Class of 2027) with hands-on experience building end-to-end machine learning pipelines across deep learning, NLP, and computer vision domains. Proficient in Python and applied ML theory including model selection, bias-variance tradeoff, statistical validity, and fairness-aware evaluation. Demonstrated ability to rapidly prototype AI solutions, manage the full ML lifecycle from data collection and labeling through model training, deployment, and maintenance. Passionate about leveraging AI to solve real-world problems, with emerging interest in HR technology and talent assessment. Strong collaborator with experience in cross-functional agile teams, technical documentation, and a curiosity-driven approach to continuous learning in fast-paced research and product environments.

Education

Bachelor of Technology in Computer Science and Engineering	2023 – 2027
Chandigarh Group of Colleges, Jhanjeri CGPA: 9.43	
Senior Secondary School Education	2022
Arya Senior Secondary School, Narwana Percentage: 91.4%	
Secondary School Education	2020
Arya Senior Secondary School, Narwana Percentage: 88.6%	

Projects

Waste AI Classification System [🔗](#)

- Led the end-to-end ML lifecycle for a computer vision classification system using CNN and deep learning with TensorFlow/Keras; managed data collection, labeling strategy, model training, hyperparameter tuning, and production deployment via Streamlit
- Benchmarked 4 deep learning architectures (VGG, ResNet, MobileNet, InceptionV3) with rigorous statistical evaluation across bias-variance tradeoff, model selection criteria, and fairness across 12 waste categories; selected InceptionV3 achieving **91% accuracy**
- Designed modular Python codebase following OOP principles; wrote technical documentation and validation tests aligned with CI/CD pipeline readiness and scalable cloud-ready architecture
- **Tech Stack:** Python, CNN, InceptionV3, TensorFlow/Keras, Deep Learning, Computer Vision, OOP, Git, Streamlit

Cryptocurrency Analyzer [🔗](#)

- Built a full-stack AI-ready data pipeline using Python Flask to fetch, parse, validate, and store live market data via REST APIs; applied statistical analysis and time-series visualization to identify market trends and anomalies
- Implemented relational data storage with MySQL/SQLite using joins, indexing, and query optimization; designed Chart.js dashboards for real-time analytics, reducing average API response latency by **35%**
- Followed end-to-end SDLC with sprint-based agile planning, Git-based collaborative version control, and modular microservices-style architecture enabling rapid prototyping of new analytical features
- **Tech Stack:** Python, Flask, JavaScript, REST APIs, MySQL, SQLite, Chart.js, Statistical Analysis, Git

Smart Travel Application [🔗](#)

- Developed a cross-platform mobile application in a 3-member agile team integrating REST APIs, Firebase, and Google Maps API; applied route optimization algorithms with computational complexity analysis for real-time geolocation processing
- Implemented clean, well-documented component-driven architecture in Flutter/Dart enabling rapid iteration and maintainability; managed source control via structured Git branching and pull-request workflows
- **Tech Stack:** Flutter, Dart, Firebase, REST APIs, Google Maps API, Optimization Algorithms, Git, Java (OOP)

Skills

- **Machine Learning & AI:** End-to-end ML lifecycle (problem scoping, data collection & labeling, model training, deployment, maintenance), Model Selection, Bias-Variance Tradeoff, Statistical Validity, Fairness-Aware Evaluation, Rapid Prototyping, Experimentation
- **AI Domains:** Deep Learning, Computer Vision (CNN, InceptionV3), Natural Language Processing (NLP), Generative AI, LLM Concepts, Retrieval-Augmented Generation (RAG), AI Integration
- **Frameworks & Libraries:** TensorFlow, Keras, Streamlit, Flask, React.js (Basics), Spring Boot (Familiar), Node.js (Basics)
- **Programming Languages:** Python (Primary), Java (OOP), JavaScript, SQL, C++, Dart
- **Data & Databases:** Data Collection, Data Labeling & Validation, MySQL, SQLite, Firebase Realtime Database, REST APIs, JSON, Relational Database Design
- **Tools & Platforms:** Git, GitHub, VS Code, Postman, Agile/Scrum, SDLC, CI/CD Basics, MongoDB Vector Search, Vertex AI
- **Core Concepts:** Data Structures & Algorithms (400+ LeetCode), Applied Statistics, OOP, DBMS, OS, Complexity Analysis
- **Soft Skills:** Research Curiosity, Technical Documentation, Cross-functional Collaboration, Fast Learning, Analytical Thinking, High Ownership, Adaptability

Achievements & Leadership

- **Ranked 1st** in Robotics and Artificial Intelligence branch — demonstrating sustained academic excellence in AI/ML domain
- **Top 700 participant across 11 countries** in the Ericsson Edge Academia Program — competitive AI/tech program with global participation
- **District Topper** in a National-Level Examination
- Campus Executive, **E-Cell IIT Bombay** — led cross-functional teams to organize technical events and workshops; strengthened collaboration, communication, and initiative
- Member, **Training Advisory Council (TAC)** — collaborated with faculty and student teams on curriculum initiatives
- Participated in **multiple hackathons** applying rapid prototyping, agile problem-solving, and AI-driven solution building under tight deadlines
- Maintained **400+ DSA Problems** solved on LeetCode with active GitHub contribution streak

Certifications

- **LLMs on Google Cloud: Retrieval-Augmented Generation on Vertex AI** – Infosys [🔗](#)
- **Introduction to Generative AI** – AWS [🔗](#)
- **Introduction to Modern AI** – Cisco [🔗](#)
- **Building AI-Powered Search with MongoDB Vector Search** – MongoDB [🔗](#)
- **Introduction to MySQL** – Infosys [🔗](#)
- **Search with MongoDB** – MongoDB Community [🔗](#)
- **Professional Skills** – IBM [🔗](#)